

Mine Planning & Economics

* Haul Road

* Clearance & Projects

* Forest clearance

* Land Acquisition

* Mine economics also deal with completion / production period revenue from the first day of mining.

* Technology advancement requires capital covered in mine planning.

* Ore — ore is aggregate of mineral which can be mined, processed & sold in profit.

* Exploration (Search & Prospecting)

(i) Resources

(ii) Reserve

Exact economically & legally

Increasing Knowledge & Confidence level ↑

→ Inferred (shape, size, depth, extent)

→ Indicate

→ Measured

Probable

Proven

Methods for ore exploration :-

(i) Drill boreholes

* Inferred has less confidence hence large spacing boreholes for cost efficient.

* Geochemical, geological & geophysical are covered in inferred.

* Indicate will have a little bit more confidence for resources hence spacing will be less.

* Mine planning

Conceptual study

Preliminary

Feasibility study

* After these studies will be use implementation.

Implementation

Procurement

Commissionary

* Processing

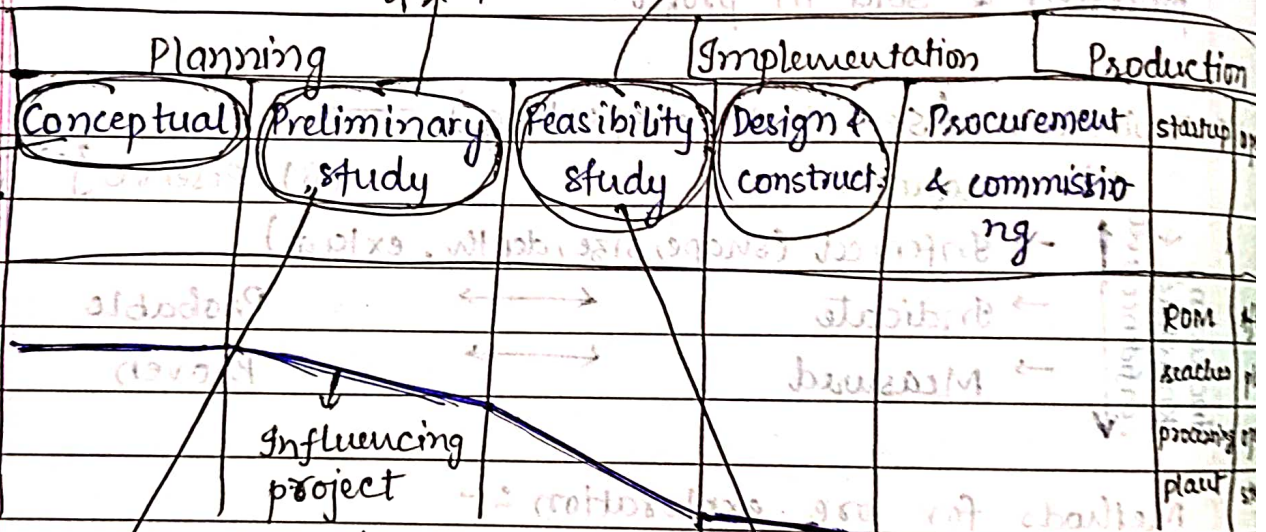
→ Gangue removing

* Production

0.3 - 0.8 % of capital cost

0.5 - 1.5 %

0.2 - 0.5 % of capital cost



Ore Tonnage of grade

Mine production & schedule

Capital cost

Operation cost

Cash flow

—

Tax

—

Revenue

→

Marketing

Profit = Revenue - Cost

→ Commercialisation

→ Reclamation

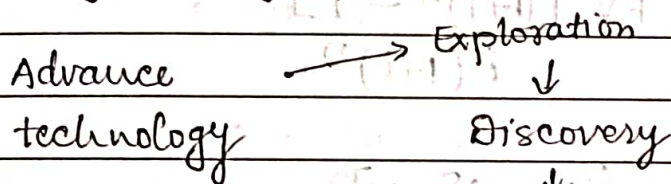
→ Metallurgy (R&D)

→ Legal ownership & laws, environment

★ Field Work check list ———

- ① Topography
- ② Climatic Conditions
- ③ Geological structures
- ④ Rock Type
- ⑤ Power supply
- ⑥ Roads
- ⑦ Surface
- ⑧ Water
- ⑨ Mine water
- ⑩ Government
- ⑪ Means of Communication

★ Mining & supply ———



(1) Delineation

★ Reclamation ———

↳ Occurrence of ore deposit
↳ Development work

★ Restoration of nature (flora & fauna)

→ Habitation

→ Rivers

↳ Mines & Process
↳ sell product

↓
Market demand

★ Mining Revenue & costs ———

↓
Changes in market

$$FV = PV(1+i)^n$$

\$1 @ 10%

$$FV(5\text{yrs}) = 1(1.1)^5 = \$1.61$$

\$5 → loan

$t=0 \rightarrow -\$5$

$t=1 \rightarrow \$1$

$\Rightarrow PV = \frac{1}{(1+i)^1}$

$t=2 \rightarrow \$1$

$\Rightarrow PV = \frac{1}{(1+i)^2}$

$\vdots$

$\Rightarrow PV = \frac{1}{(1+i)^2}$

$t=5 \rightarrow \$1$

$\Rightarrow PV = \frac{1}{(1+i)^5}$

$r = 10\%$

$\Rightarrow PV = \frac{1}{(1+i)^3}$

$$S = \frac{a(1-r^n)}{(1-r)}$$

$$PV = \frac{1}{(1+i)^5}$$

$$NPV = \frac{FV}{1+i} \left[1 - \left(\frac{1}{1+i} \right)^n \right]$$

$$\left(\frac{1 - \frac{1}{1+i}}{1+i} \right)$$

$$= \frac{FV [(1+i)^n - 1]}{i(1+i)^n}$$

$$NPV = \frac{1 [(1+0.1)^5 - 1]}{0.1(1.1)}$$

$$= 3.78$$

$$\frac{FV [(1+i)^n - 1]}{i(1+i)^n} = 5$$

$$n \approx 7 \text{ yrs}$$

30/07/24

- * Payback Period
- * Cashflow
- * Discounted cashflow (DCF)
- * Discounted cashflow rate of return (DCFROR)
- * Cashflow, DCF & DCFROR including depreciation
- * Cashflow, DCF & DCFROR including depletion

\$5 → loan

$t=0 \rightarrow -\$5$

$t=1 \rightarrow \$1 \Rightarrow PV = \frac{1}{(1+i)^1}$

$t=2 \rightarrow \$1$

$\vdots$

$\Rightarrow PV = \frac{1}{(1+i)^2}$

$t=5 \rightarrow \$1$

$r = 10\%$

$\Rightarrow PV = \frac{1}{(1+i)^3}$

$\vdots$

$PV = \frac{1}{(1+i)^5}$

$$S = \frac{a(1-r^n)}{(1-r)}$$

$$NPV = \frac{FV}{1+i} \left[1 - \left(\frac{1}{1+i} \right)^n \right]$$

$$\left(\frac{1 - \frac{1}{1+i}}{1+i} \right)$$

$$= \frac{FV [(1+i)^n - 1]}{i(1+i)^n}$$

$$NPV = \frac{1 [(1+0.1)^5 - 1]}{0.1(1.1)}$$

$= 3.78$

$$\frac{FV [(1+i)^n - 1]}{i(1+i)^n}$$

$= 5$

$n \approx 7 \text{ yrs}$

30/07/24

- * Payback Period
- * Cashflow
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$$5 = 1 \left(\frac{(1.1)^n - 1}{0.1(1.1)^n} \right) = \frac{1}{0.1} - \frac{1}{0.1(1.1)^n}$$

$$\Rightarrow \frac{10^2}{(1.1)^n} = 10 - 5 = 5$$

$$\Rightarrow (1.1)^n = 2$$

$$\Rightarrow n \approx 7.3 \text{ yrs}$$

* Cashflow

Gross revenue

- operating expense

= Taxable Income

- Tax

= Profit

- Capital cost

= cashflow

$$Q. \text{ NPV} = -200 - \frac{100}{(1.1)^1} + \frac{100}{(1.1)^2} + \frac{100}{(1.1)^3} + \frac{120}{(1.1)^4} + \frac{130}{(1.1)^5} + \frac{140}{(1.1)^6}$$

$$= 116.087$$

$$= 116.087$$

* Salvage value — Garbage value of non-usable machinery by selling it.

We consider it is 0 - but in reality its not.

* Depreciation

Project life = 10 yrs

Capital cost = 600

Gross Revenue

- Operating Cost

- Depreciation

= Taxable Income

- Tax

= Profit

- Capital cost

+ Depreciation = cashflow

★ Depletion ———— $(1 - (1 - 0.1)^{10}) \times 1 = 0.614$
 → Depreciation of reserve like depreciation of investment

- (i) cost basis depletion deduction
- (ii) Percentage depletion deduction

cost basis = property acquisition + exploration + geophysical + geochemical + geological study

Total Reserve = 100 tons
 Yearly = 10 tons

cost basis = \$10

$$\frac{\$10}{100} = 0.1 \$ / \text{ton}$$

After 10 yrs = \$0.1 / ton $\times 10$

$$100 - 10 \times 10 = 0$$

Gross revenue

- operating cost

- Depreciation

- Depletion

= Taxable Income

= Profit

- Capital cost

+ Depletion

+ Depreciation

= Cash flow

Q. Interest rate = 15%

$$NPV = -100 + \frac{40.4}{(1.15)^1} + \frac{41.8}{(1.15)^2} + \frac{43.1}{(1.15)^3} + \frac{44.5}{(1.15)^4} + \frac{45.8}{(1.15)^5}$$

$$= 43.28$$

$$0 = NPV$$

$$0 = -100 + \frac{40.4}{(1+i)} + \frac{41.8}{(1+i)^2} + \frac{43.1}{(1+i)^3} + \frac{44.5}{(1+i)^4} + \frac{45.8}{(1+i)^5}$$

$$(1+i) = 1.315$$

$$(1+i) = 1.315$$

$$i = 0.315$$

$$ROR = 31.5\%$$

Q. Capital @ 5% compounded daily

Effective simple interest rate

Project cash flow	0	1	2	3	4	5	6
	-200	-100	+100	+110	+120	+130	+140

$$CI = A - P$$

$$= P \left(1 + \frac{r}{n} \right)^n - P$$

$$= P \left(1 + \frac{0.05}{365} \right)^{365} - P$$

$$= P \left[\left(1 + \frac{0.05}{365} \right)^{365} - 1 \right]$$

$$= 0.5126P$$

POS - Point of sale

FOB - Free on Board

CIF - Cost Insurance Freight

★ Revenue Estimate -

→ Current Mineral Price

(i) Newspaper

(ii) Mining magazine

(iii) Metals week

(iv) Mining Bulletin

} present prices & based on these forecast the prices are also done

★ Price depends upon -

→ quality

→ quantity

→ source

→ form

→ packaging

Units

long ton

short ton

1 metric ton = 1000 kg

1 st = 0.901 mt

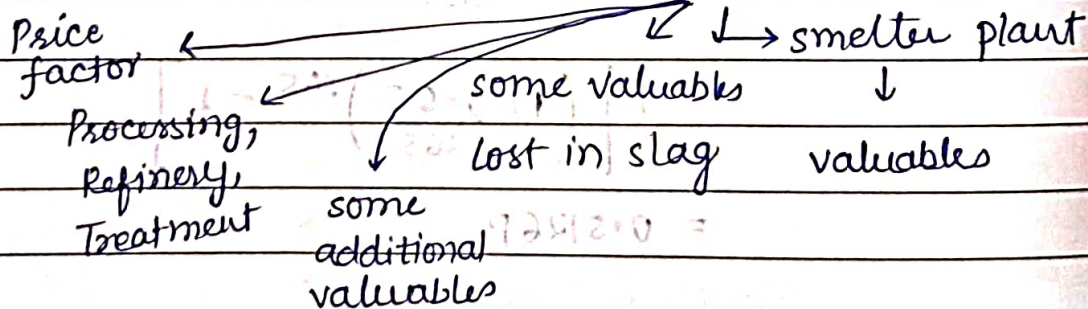
1 lt = 1.06 mt

1 troy ounce = 33/g

★ In FOB, all charges have to pay (transportation, production, etc.)

Market → Cu, Pb, Zn → refined product

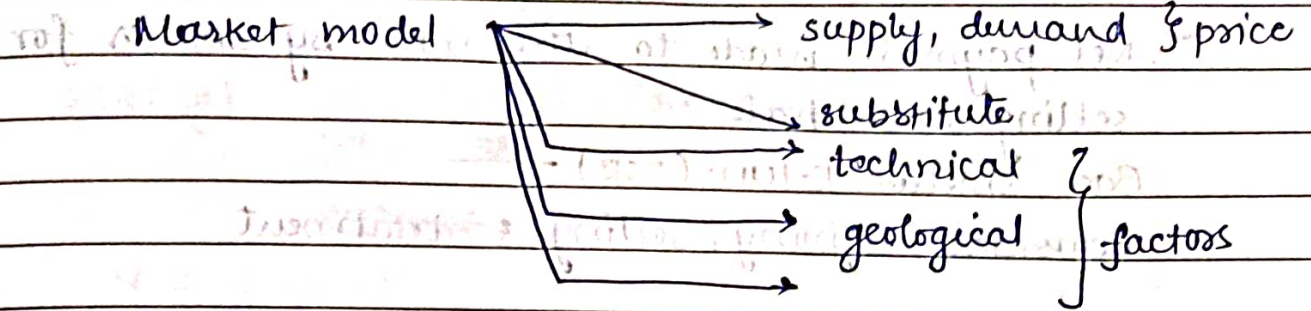
some minerals sold as concentrate



(1) Trend Analysis -

with past prices we plot a curve to forecast the future prices.

(2) Econometric model —→



* Net smelter ~~per~~ ~~ton~~ Run —→

1 ton —→ G % grade { 1 ton = 2204.623 pounds }

$$\text{Metal content (lb)} = \frac{G}{100} \times 2000$$

* some variables lost in slag

→ unit
→ percentage

{ 1 ton = 1000 kg }

{ 1 lb = 0.45 kg }

$$1 \text{ ton} = 1000 \times \frac{1}{0.45} = 2222.2 \text{ lb}$$

$$\text{Me (lb)} = \frac{C}{100} \times \left(\frac{G\text{-unit}}{100} \right) \times 2000$$

{ Me = Metal content
C = Metal credit }

$$G_e = \frac{C}{100} \times G\text{-unit}$$

$$\text{Price} = \text{Me} \cdot P \quad \{ P = \text{cost / lb} \}$$

* factor also comes into picture where prices lost in slag, processing...

$$\text{Price} = \text{Me} \cdot P \cdot f \quad \{ f = \text{price factor} \}$$

$$GV = \text{Me} \cdot P \cdot f - X + Y - T$$

where; X = excess is unwanted mineral

Y = valuable occurring with ~~concentrate~~ concentrate /

mineral purchase

T = Treatment charge

$$GV = \text{Me} \cdot (P \cdot f - r) - X + Y - T$$

$\text{¢} = \text{cent}$

Net smelter Return (NSR) —

→ Net payment made to the mine by smelter for selling concentrate

Base smelter Return (BSR) —

→ Considers refining, selling & treatment

$$CM \text{ (Metal content)} = \frac{G}{100} \times 2000$$

$$CV \text{ (contained value)} = \frac{G}{100} \times 2000 (P) \rightarrow \$/\text{lb}$$

↳ \$/ton

$$\text{Payment \% (PP)} = \frac{AMR}{CV} \times 100$$

$$G_e = \frac{C(G-u)}{100} \times 2000$$

$$M_e = \frac{C(G-u)}{100 \times 1000} \times 2000$$

$$BSR = M_e(P.f - r) - T$$

$$NSR = M_e(P.f - r) - T - X + Y$$

$$AMR = NSR - R$$

→ Estimated
Realisation cost

(freight, insurance, weighing)

Q. Copper concentrate (G) = 30%

~~if~~ u = 1%

$$BSR = \frac{98 \times (30-1)}{100} \times \frac{2000}{100} \times \left[\$1 \times 1 - 0.1 \right] - \$75$$

$$= \$436.56$$



$$NSR = \frac{98}{100} \times \frac{(30-1)}{100} \times 2000 \times [\$1 \times 1 - 0.1] - \$75 - x + y$$

$$BSR(Ag) = \frac{95}{100} \times \frac{(30-1)}{100} \times (\$6 \times 1 - 0.35)$$

$$= \cancel{\$181.95} \quad \$155.65$$

$$y = \$164.95$$

$$CIX = (2-1) \$10 = 10$$

$$NSR = \$436.58 - \$10 + \cancel{\$164.95} + \$155.65$$

$$= \$582.21$$

$$AMR = NSR - R \quad \text{ev} = \frac{98}{100} \times 2000$$

$$= \$582.21 - 0.39 R_m^{0.7}$$

$$= \$551.09$$

$$PP = \frac{AMR}{1960} \times 100$$

$$= \frac{551.09}{1960} \times 100$$

$$= 28.11\%$$

$$R = 0.39 \times R_m^{0.7}$$

$$CV = \frac{30 \times 2000 \times \$ (30 \times 6)}{100} + \frac{2 \times 0.5 \times 2000}{100}$$

$$\cancel{\$108000} + \cancel{\$2000} = 800$$

$$\cancel{\$108020}$$

$$PP = \frac{551.09}{800} \times 100$$

$$= 69.026\%$$

$$NSR = M_e (P_f - r \pm e_1) - (T \pm e_2) - x + y$$

escalation
selling
refining

escalation
in treatment

★ Price-cost relationships —

T → treatment cost

P → price

r → refining & selling cost

$$BSR = \frac{100}{100} \times \frac{(28.5 - 1.3)}{100} \times 2000 \times \$ (90 \times 1 - 12) - \$60$$

$$= \$364.32$$

Assume that next year both the price and the costs increase by 5% but that c, G and u remain constant

$$P = 90 \times 1.05 = 94.5$$

$$T = 60 \times 1.05 = \$63$$

$$r = 12 \times 1.05 = \$12.6$$

$$NSR = 5.44 \times 81.9 - 63$$

$$= \$382.54$$

Explosive
Drill bits
Detonator

powers

Driller, helper

Land & equipment

★ Estimating cost —

① Capital cost → ownership cost

② Operational cost → production cost (Drilling, blasting, loading, hauling, grading, others)

③ General & Administration cost

Material
Labour

Direct

Indirect (conveyor, supervision)

07/08/2024

★ O'Hara Cost Estimator —

t = tons of ore milled/day

t_o = tons of ore mined/day

t_w = tons of waste minerals/day

t_c = tons of ore crushed/day

t_p = t_o + t_w

mill plant runs on all 7 days of a week whereas mining activity is carried out usually 5 days in a week.

* Milling plant must have ore available on no-mining day.

5-6 months
concentrate
upto

$$5 t_o = 7 t$$

$$\Rightarrow t = \frac{5 t_o}{7} = 0.71 t_o$$

$$t_o = \frac{7 t}{5} = 1.4 t$$

$$\text{Productivity} = \frac{\text{ore-mined}}{\text{manshift}} = \frac{10,000}{100} = 1000 \text{ tons/manshift}$$

Nop (Number of personnel in open pit mines for shovel & dumping)

Nml (for mill plant)

Tp (ore & waste both required)

T (ore is ~~ore~~ required only)

09/08/24 Estimating costs → Detailed cost calculation

Basic equipments

- Shovels
- Dumpers/Trucks
- Drills

Sum

- Depreciation

Q. 3M tons $\rightarrow$ ore 4 2M tons $\rightarrow$ waste

Mine will operate 2 shifts/day, 5 days/week, 52 weeks/year, with 10 holidays.

per shift : ore = 6000 tons

waste = 4000 tons

★ Production capacity / Number of machines

Drills : 35 ft of hole per hour

shovels (waste) : 630 tons / hour

shovels (ore) : 750 tons / hour

Trucks (waste) : 175 tons / hour

Trucks (ore) : 280 tons / hour

ore : —

1 drill (1 shift / day), 1 shovel (2 shifts / day),
3 trucks (2 shifts / day)

waste : —

1 drill (1 shift / day), 1 shovel (2 shifts / day),
4 trucks (2 shifts / day)

MEP — Material Expenses

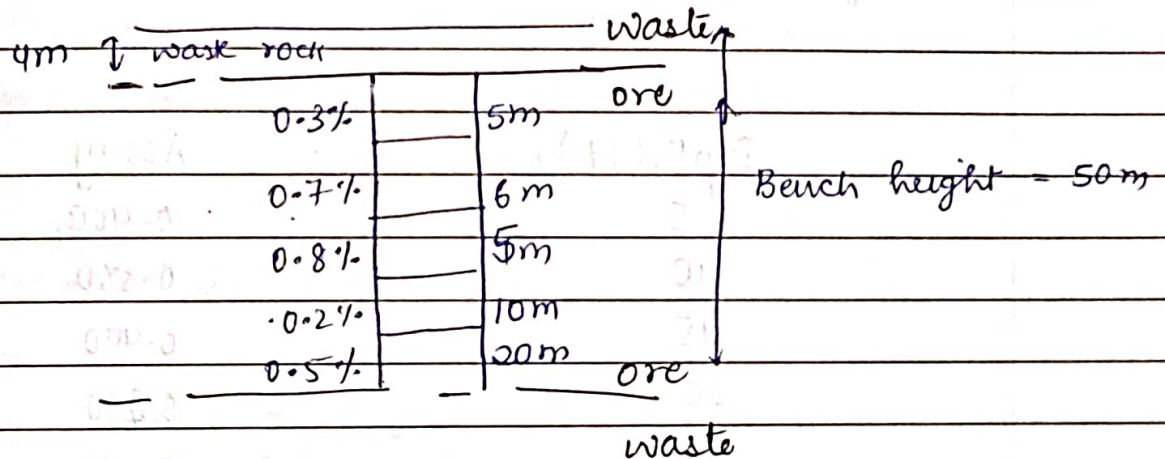
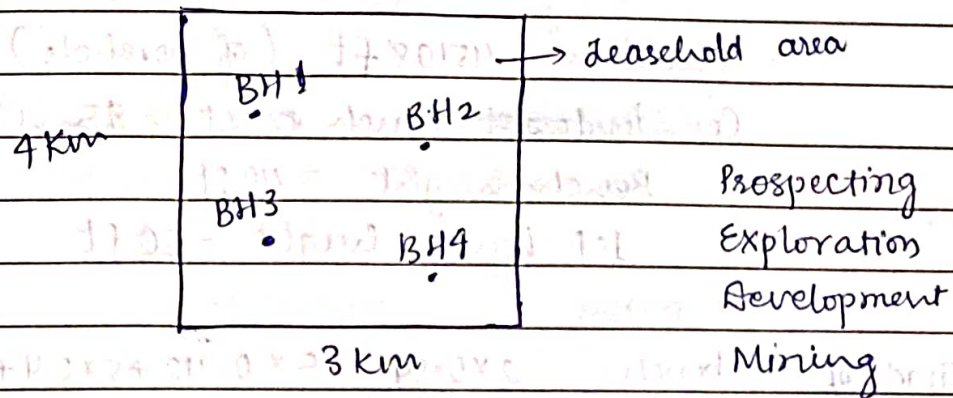
size, shape, area, volume, mineral content

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★ Orebody Description —

- Borehole ~~Setting~~ Lithology sections → Logging ↓
- Mine plane → scale → Assaying cores recovered
- Compositing → Mineral Content → RQD
- Tonnage factor (TF) → physico-mechanic Properties



$$\frac{5 \times 0.3 + 6 \times 0.7 + 5 \times 0.8 + \dots + 20 \times 0.5}{5 + 6 + 5 + 10 + 20}$$

Mine plan → Geological plan
→ surface plan

$$\text{scale} = \frac{\text{Distance on plan}}{\text{Distance in actual/measure}}$$

$$n \times 0.01 \times 0.5 = 5000$$

$$n = 10^6 \text{ tons}$$

★ Compositing & Tonnage Factor calculations —

compositing = Average grade

$$\Rightarrow \bar{g} = \frac{\sum l_i g_i}{\sum l_i} \rightarrow \text{Bench height}$$

Elevation = 45198 ft (of borehole)

Considered ~~rest~~ bench crest = 45200 ft

Bench height = 40 ft

1st bench height = 30 ft

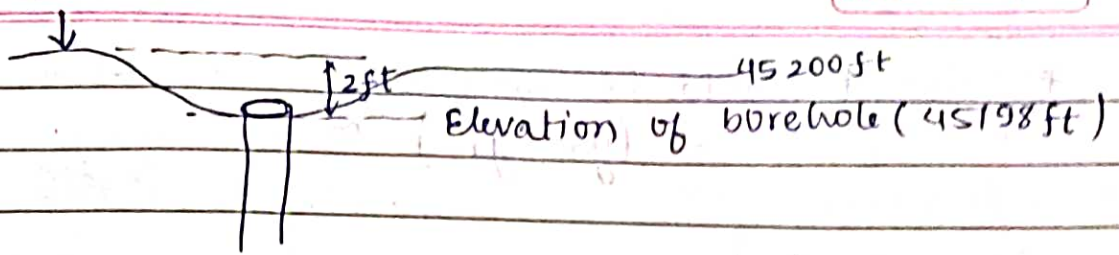
Grade of 1st bench = $2 \times 0.59 + 5 \times 0.48 + 5 \times 0.4 + 5 \times 0.38 + \dots + s \times$

Depth (ft)	Assay
5	0.400
10	0.560
15	0.440
20	0.480
25	0.400
30	0.380
35	0.380
40	0.590
45	0.480
50	0.600
55	0.560
60	0.320
65	0.700
70	
75	
80	
85	

considered bench crest

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$$\text{Buoyancy} = \rho_b \cdot V \cdot g$$

$$\text{Tonnage factor} = \frac{1}{\rho} \times \frac{V}{m}$$

* Estimating reserve — (Geostatistics)

Inverse Distance

Methods

Horizontal section

Triangle

Polygon

vertical section

Kriging

Pit-layout rule —

* Bottom of the bench end in low grade ore.

* ~~Bench~~ Bench angle $\rightarrow 27^\circ \rightarrow$ soft (surface)

$54^\circ \rightarrow$ stone rock

41°

20/08/24

(g₀)

* unknown grade at (x₀, y₀)

$$g_0 = \sum_{i=1}^n a_i g_i$$

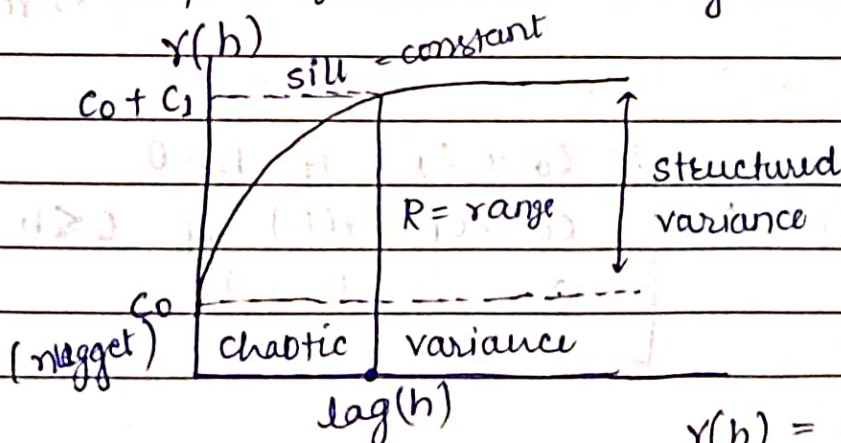
$$a_i = \frac{\frac{1}{d_i^m}}{\sum_{i=1}^n \frac{1}{d_i^m}}$$

m = 2

$$\sum a_i = 1$$

* Variogram — plot of variance vs lag distance

spherical model



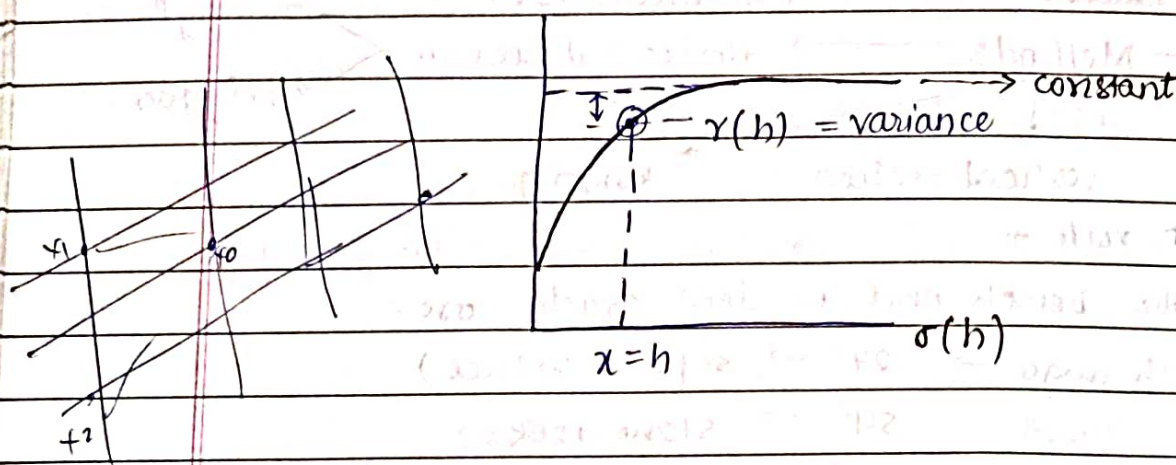
$$r(h) = \frac{1}{n(h)} \sum (g_i - g_j)^2$$

* nugget should be minimum.

$$\rightarrow (g_i - g_j) \approx 0$$

$$\gamma(h) = \frac{1}{2n(h)} \sum (g_i - g_j)^2$$

* Covariance



$(\sigma(h))$ covariance = sill - variance

$$= C_0 + C_1 - \gamma(h)$$

$$\gamma(h) = \begin{cases} 0 & \text{if } h = 0 \end{cases}$$

$$C_1 \left(\frac{3h}{2a} - \frac{1}{2} \left(\frac{h}{a} \right)^3 \right) + C_0 \quad \text{if } 0 \leq h \leq a$$

$$C_0 + C_1$$

if $h > a$

range

$$\sigma(h) = \begin{cases} C_0 + C_1 & \text{if } h = 0 \\ C_0 + C_1 - \gamma(h) & \text{if } 0 \leq h \leq a \\ 0 & \text{if } h > a \end{cases}$$

Model
 Linear ← { } → spherical
 Dewijzen

Model is fitted

* Kriging Technique

lower & Dewijzen are good models for accurate at lower values of lag distance

$$g_0 = \sum a_i g_i$$

$$= a_1 g_1 + a_2 g_2 + \dots + a_n g_n$$

$$a_1 = \frac{1}{d_1^2}$$

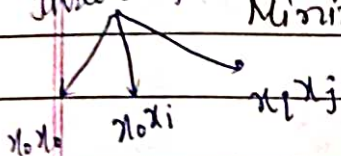
$$\frac{1}{d_1^2} + \frac{1}{d_2^2} + \dots + \frac{1}{d_n^2}$$

$$= \frac{1}{d_1^2}$$

$$\sum_{i=1}^n d_i^2$$

(Total variance) $\sigma_e^2 = \sigma_{x_0}^2 - 2 \sum_{i=1}^n a_i \sigma_{x_0} x_i + \sum_{i=1}^n \sum_{j=1}^n a_i a_j \sigma_{x_i} x_j$

Three components



Minimum then correct g_0 is estimated

For minimum value, partial differentiation of total variance $(\sigma_e)^2$.

$$-\sum_{i=1}^n \sigma_{x_0} x_i + \sum_{j=1}^n a_j \sigma_{x_i} x_j = 0$$

using Lagrange's multiplier...

weighting coefficient

$$\sum_{j=1}^n a_j x_i x_j + \lambda = \sigma_{x_0} x_i$$

→ n equations

so to solve we need $(n+1)$ equations

$(n+1)^{th}$ equation is : $\sum a_i = 1$

$$h_{01} = 141$$

$$h_{02} = 141$$

$$h_{03} = 224$$

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$$\eta(h) = \eta - 1$$

$$\eta(2h) = \eta - 2$$

$$\eta(3h) = \eta - 3$$

$$\vdots$$

$$\eta(kh) = \eta - k$$

$$r(h) = \frac{1}{\eta h} \left[\sum (g_i - g_j)^2 \right]$$

Example —

$$x_1 \quad C-37 \quad g_1 = 0.320$$

$$x_2 \quad C-25 \quad g_2 = 0.230$$

$$x_3 \quad C-26 \quad g_3 = 0.833$$

$$(i) \quad g_0 = a_1(0.320) + a_2(0.230) + a_3(0.833)$$

$$(ii) \quad a_1 + a_2 + a_3 = 1$$

$$(iii) \quad \sum_{j=1}^n a_j \sigma_{x_i, x_j} + \lambda = \sigma_{x_0, x_i}$$

$$a_1 \sigma_{11} + a_2 \sigma_{12} + a_3 \sigma_{13} + \lambda = \sigma_{01} \quad \text{--- (i)}$$

$$a_1 \sigma_{21} + a_2 \sigma_{22} + a_3 \sigma_{23} + \lambda = \sigma_{02} \quad \text{--- (ii)}$$

$$a_1 \sigma_{31} + a_2 \sigma_{32} + a_3 \sigma_{33} + \lambda = \sigma_{03} \quad \text{--- (iii)}$$

$$a_1 + a_2 + a_3 = 1 \quad \text{--- (iv)}$$

$$h_{00} = 0 \quad h_{11} = 0 \quad h_{22} = 0 \quad h_{33} = 0$$

$$h_{01} = 141 \quad h_{02} = 141 \quad h_{03} = 224$$

$$h_{12} = 200 \quad h_{13} = 361$$

$$h_{23} = 300$$

$$r = \begin{cases} 0 & \text{if } h = 0 \\ C_1 \left(\frac{3h}{2a} - \frac{1}{2} \left(\frac{h}{a} \right)^3 \right) + C_0 & \text{if } 0 < h \leq a \\ C_1 + C_0 & \text{if } h > a \end{cases}$$

$$a = 450$$

$$C_0 = 0.02$$

$$C_1 = 0.16$$

$$\text{sil} = 0.18 = C_0 + C_1$$

$$\text{Negget}(C_0) = 0.02$$

$$a_1 = 0.389$$

$$a_2 = 0.3584$$

$$a_3 = 0.2517$$

$$\sigma_{11} = 0.18$$

$$\sigma_{12} =$$

$$\sigma_{13} =$$

$$\sigma_{22} = 0.18$$

$$\sigma_{23} =$$

$$\sigma_{33} = 0.18$$

$$\sigma_e^2 = \underbrace{\sigma_{x_0 x_0}}_{C_0 + C_1 = \text{sil}} - 2 \sum_{i=1}^n a_i \sigma_{x_0 x_i} + \sum_{i=1}^n \sum_{j=1}^n a_i a_j \sigma_{x_i x_j}$$

$$\sigma_e^2 = \sigma_{x_0 x_0} - \sum a_i \sigma_{x_0 x_i} - \lambda$$

$$\lambda = -0.0067$$

$$\begin{aligned} \sigma_e^2 &= 0.18 - (a_1 \sigma_{01} + a_2 \sigma_{02} + a_3 \sigma_{03}) - 0.0067 \\ &= 0.1087 \end{aligned}$$

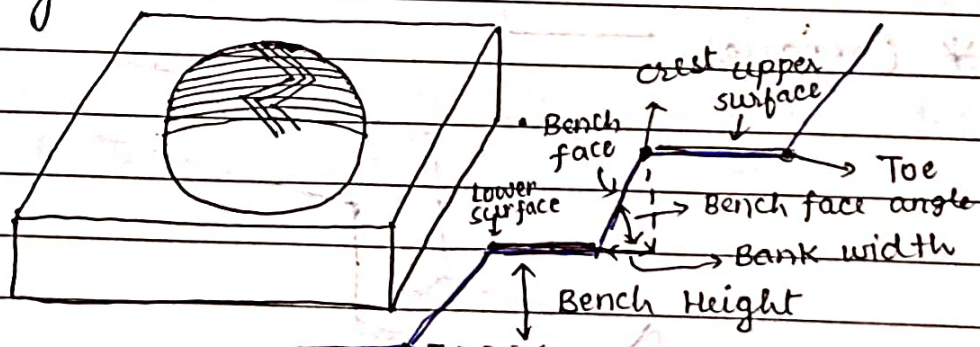
$$\text{standard deviation} = \sqrt{\sigma_e^2}$$

$$= \sqrt{0.1087}$$

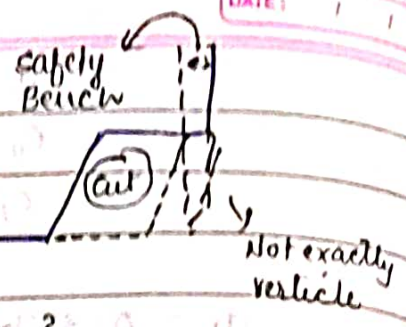
$$= 0.329$$

3/08

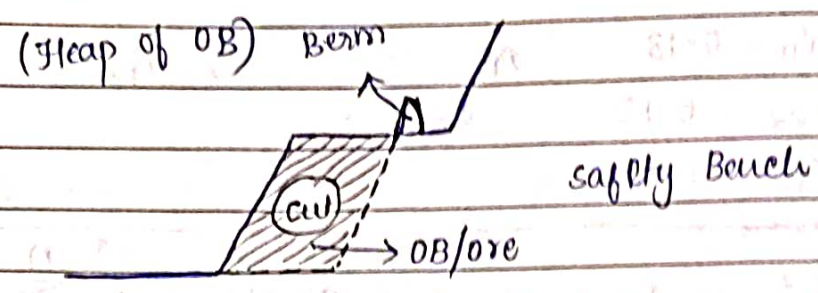
★ Bench Geometry —



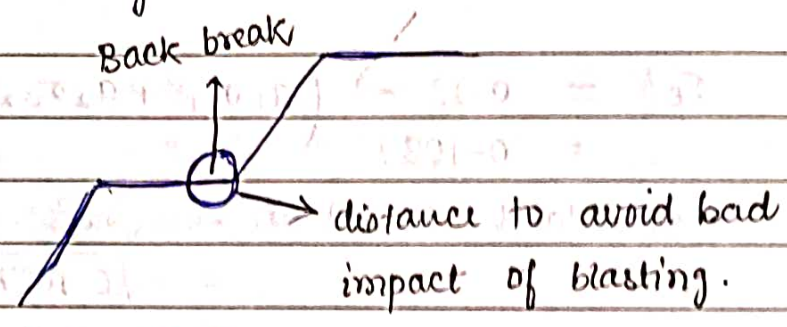
- * Berms
- * safety Bench / catch Bench



Q) why safety Bench / catch Bench is left?

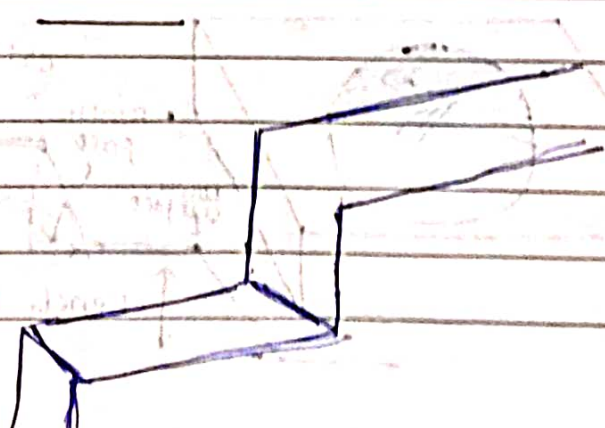


- * Earlier bench height used to be decided by drilling equipment changes to modern day loading equipment (shovel).
- * Bench geometry is very important for production.
- * Optimal bench width is very important as narrow bench width may lead to toppling of vehicle & wider bench may lead to dilution.



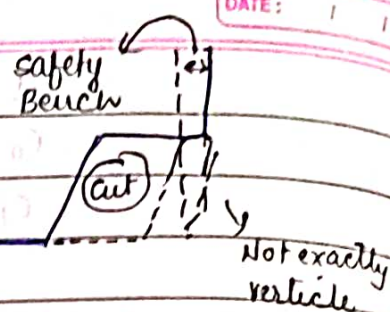
- * Bench crest
- * Design crest

* Orepasses



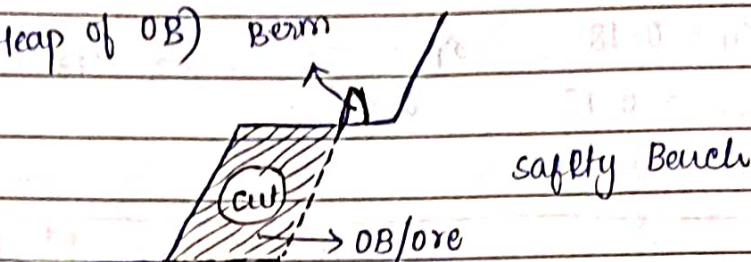
* Berms

* Safety Bench / catch Bench



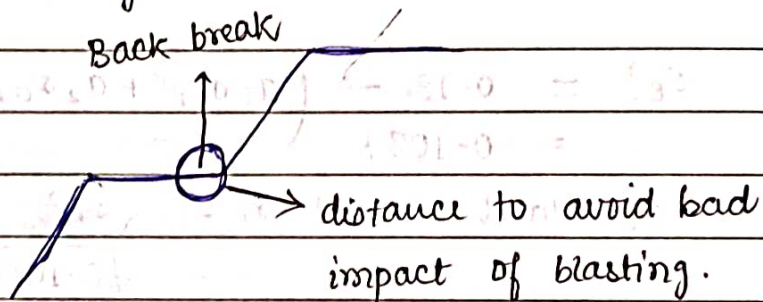
Q. why safety Bench / catch Bench is left?

(Heap of OB) Berm



- * Earlier bench height used to be decided by drilling equipment changes to modern day loading equipment (shovel).
- * Bench geometry is very important for production.
- * Optimal bench width is very important as narrow bench width may lead to toppling of vehicle & wider bench may lead to dilution.

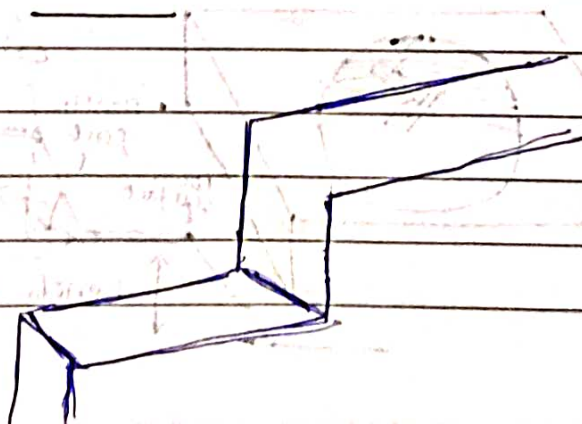
Back break



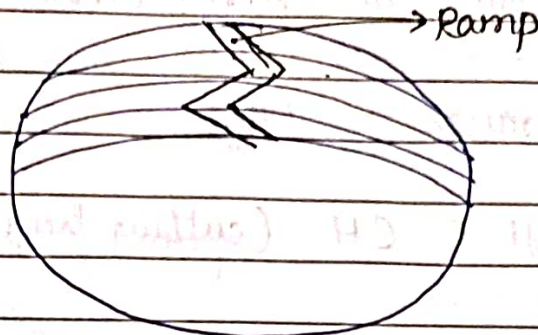
* Bench crest

* Design crest

* Orepasses

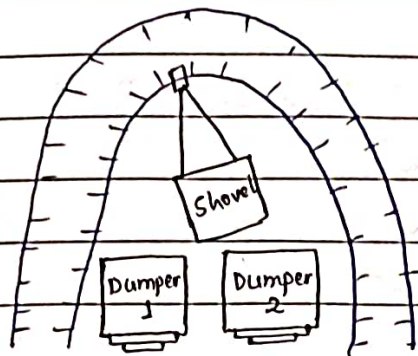


- * Ramps are for covering shortest possible distance from depth to surface maintaining optimal gradient.



27/08

Bench Width



single spotting
double spotting

Radius of turning
of excavator = height

Bench 1

Bench 2

Bench 3

Drilling & Blasting

Cut-1

Cut-2

Cut-3

single spotting (waiting time of shovel)
↓

Idle timing of shovel (space constraint)

- * Both too width bench & narrow width bench has pros & cons.

* No waiting time for shovel in parallel-cut.

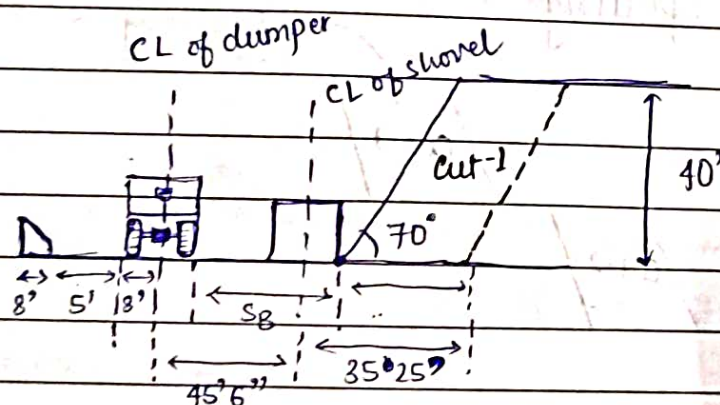
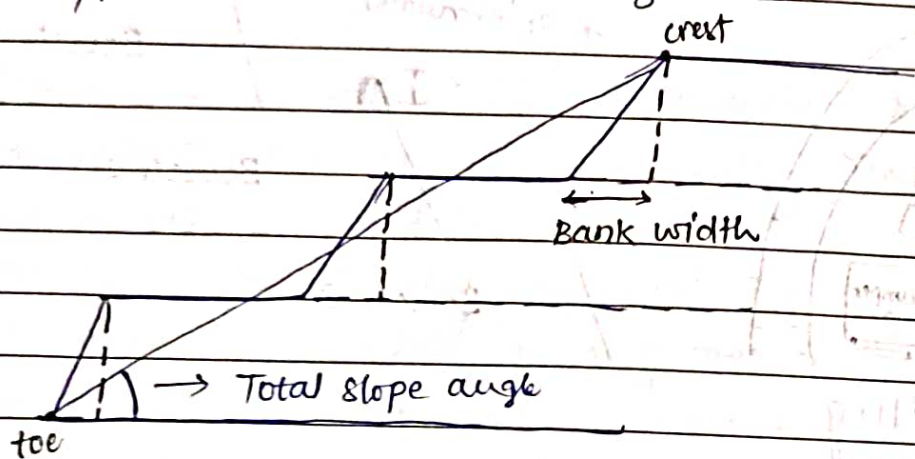
* Double spotting has no waiting time.

* Shovel working range - (Page

(i)

(Bench height) $BH < CH$ (cutting height)

* Total slope angle = line joining the lower most toe & uppermost crest to horizontal.



Bench width $\approx 102'$

Safety bench width $\rightarrow$

Q. > what is the distance between CL of dumper & shovel?

h

* Pit Expansion

(i) Frontal cut

(ii) Parallel cut - drive by

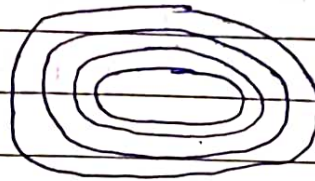
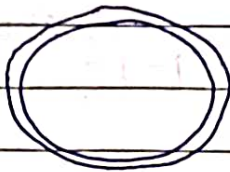
(iii) Parallel cut - turn & back

→ sufficient space is there

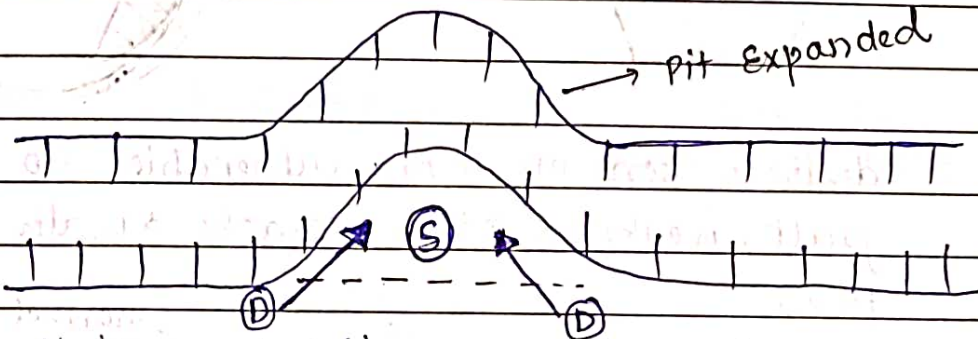
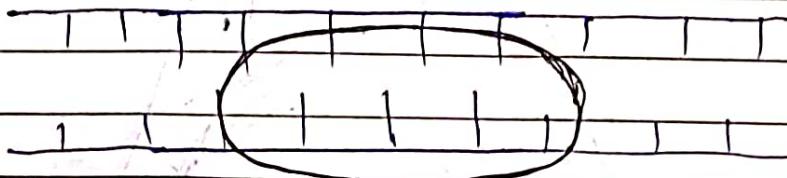
→ narrow benches

* Ore Access

→ Formation of ramps

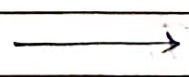
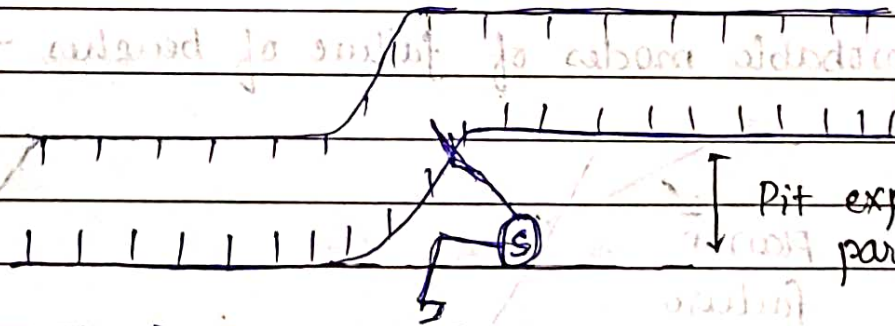


Frontal cut

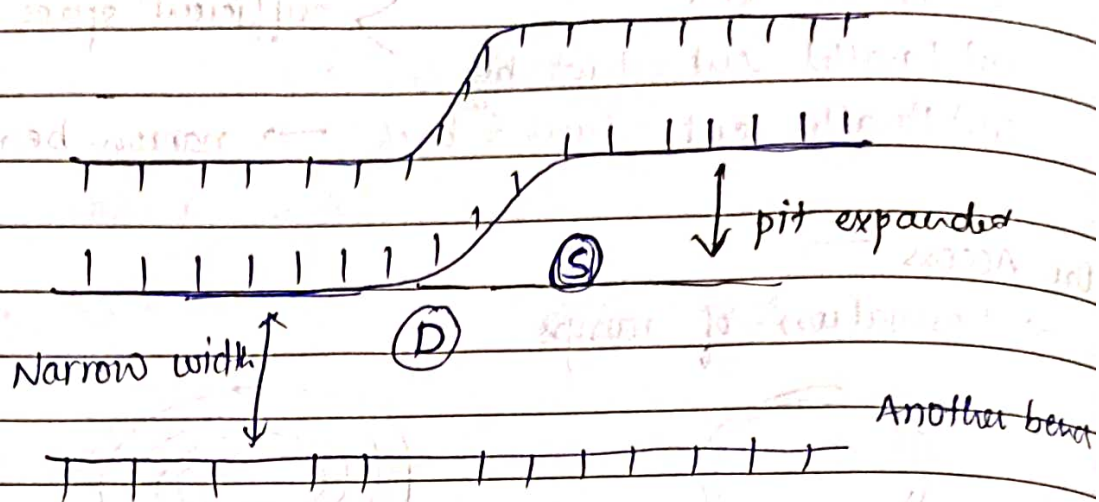


* Minimum operating room to work on a bench

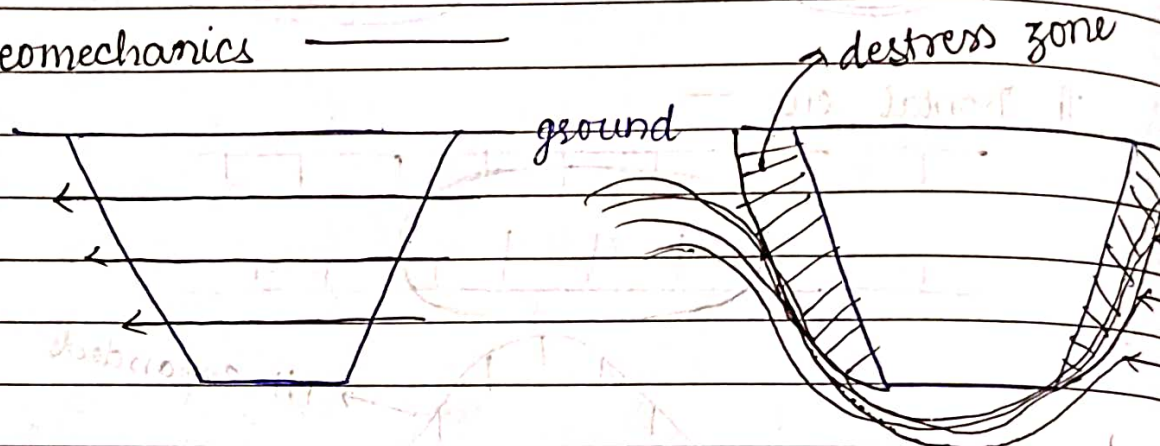
Parallel cut - drive by



Parallel cut - turn & back



* Geomechanics

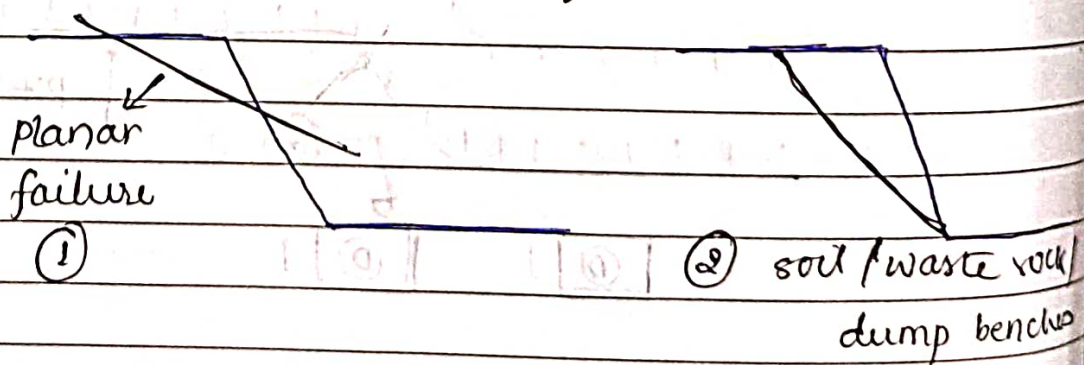


distress zone are more vulnerable to joint, fault, weakness plane, cracks → water can easily flow.

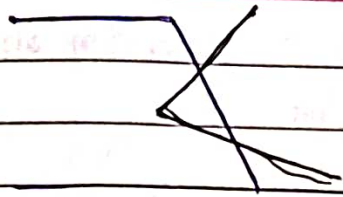
* for safety of benches: $\tau = c + \sigma \tan \phi$

normal ↑
shear ↑ cohesion internal angle b/w friction plane

* 4 probable modes of failure of benches



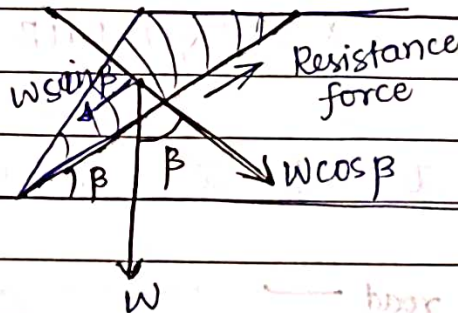
(3)



(4)



* circular type failure (e.g. dump benches) are more prominent because they are fragmented.



$$\tau = c + \sigma \tan \phi$$

$$\tau_A = c_A + \sigma_A \tan \phi$$

Driving Resisting

$$\text{Factor of safety (F)} = \frac{\text{Resisting force}}{\text{Driving force}}$$

$$F = \frac{c_A + W \cos \beta \tan \phi}{W \sin \beta}$$

(dry)

$$\text{(wet)} \quad F = \frac{c_A + (W \cos \beta - U) \tan \phi}{(W \sin \beta + V)}$$

vertical stress

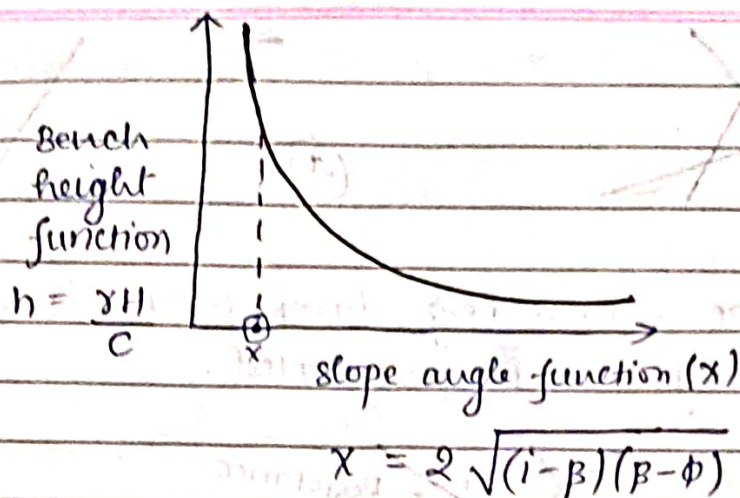
$$\sigma_v = \gamma H$$

Bench height

$$FOS = 1$$

$$\gamma H = c \Rightarrow h = \frac{\gamma H}{c}$$

$$\left\{ \begin{array}{l} H = 500 \text{ ft} \\ \gamma = 160 \text{ lb/ft}^3 \end{array} \right.$$



Example -

$$\beta = 50^\circ \quad \phi = 30^\circ \quad \chi = 17^\circ$$

30/08/24

★ Addition of haul road —

- ① Design of spiral road - inside pit wall
- ② Design of spiral road - ~~outside~~ outside pit wall

Let Bench Height = 30'

width = 30'

crest - crest distance = 60'

Bench face angle = 56°

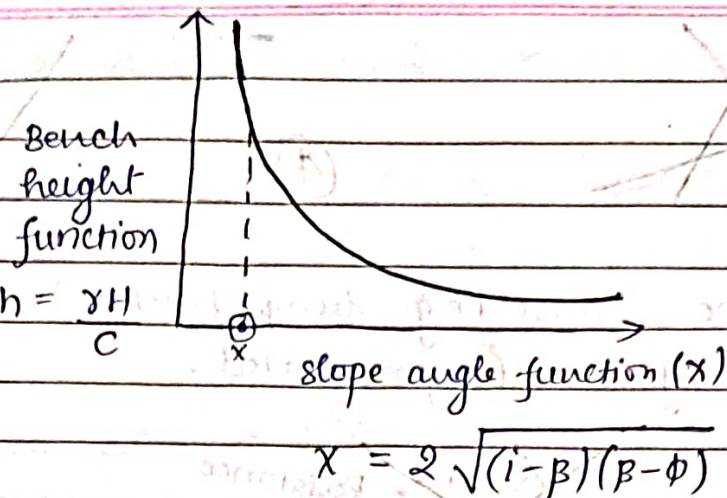
Road gradient = 10%

laying road

- sub-grade
- sub base
- base
- wearing surface

Road - Lane

- spiral
- traffic direction
- switchbacks
- gradient = 8% - 10%



Example -

$$\beta = 50^\circ \quad \phi = 30^\circ \quad \chi = 17^\circ$$

30/08/24

★ Addition of haul road —

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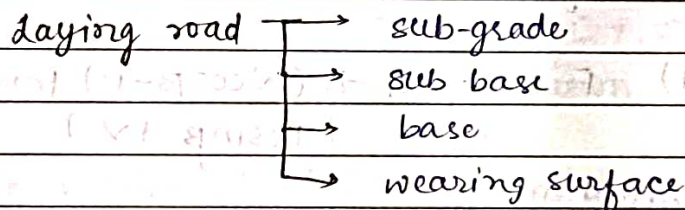
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Road - Lane

- spiral
- traffic direction
- switchbacks
- gradient = 8% - 10%